Distribution of Accessibility Scores (Tiered E2SFCA)

The SURVEYFREQ Procedure

Data Summary	
Number of Strata	18
Number of Clusters	771
Number of Observations	1500
Number of Observations Used	777
Number of Obs with Nonpositive Weights	723
Sum of Weights	793.858

Table of colon_screening by screening_behavior

colon_screening	screening_behavior	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
.	.	2	1.20400	0.85441	0.1517	0.1078	18.1708	14.0028
	1	1	0.60200	0.60200	0.0758	0.0759	9.0854	9.4550
	2	4	4.82000	3.25551	0.6072	0.4091	72.7437	17.7224
	3	0	.	.	.	.	.	.
	4	0	.	.	.	.	.	.
	Total	7	6.62600	3.40919	0.8347	0.4285	100.0000	
1	.	6	3.47500	1.54912	0.4377	0.1959	1.1569	0.5181
	1	129	124.55200	13.05614	15.6895	1.5905	41.4648	3.4588
	2	122	114.76800	12.17981	14.4570	1.5040	38.2076	3.3778
	3	35	32.41400	6.71431	4.0831	0.8407	10.7910	2.1517
	4	24	25.17100	6.08435	3.1707	0.7639	8.3797	1.9579
	Total	316	300.38000	18.09214	37.8380	2.1050	100.0000	
2	.	22	16.13700	3.79904	2.0327	0.4827	3.3146	0.7862
	1	163	166.99600	14.90060	21.0360	1.8005	34.3012	2.7203
	2	166	175.59800	15.58529	22.1196	1.8656	36.0680	2.7757
	3	63	71.83800	10.67849	9.0492	1.3166	14.7556	2.0787
	4	40	56.28300	10.45326	7.0898	1.2823	11.5606	2.0314
	Total	454	486.85200	21.77799	61.3273	2.1215	100.0000	
Total	.	30	20.81600	4.16922	2.6221	0.5325		
	1	293	292.15000	18.55413	36.8013	2.1361		
	2	292	295.18600	18.63719	37.1837	2.1439		
	3	98	104.25200	12.44849	13.1323	1.5235		
	4	64	81.45400	11.92432	10.2605	1.4540		
	Total	777	793.85800	21.07053	100.0000			

Distribution of Accessibility Scores (Tiered E2SFCA)

The SURVEYFREQ Procedure

Data Summary	
Number of Strata	18
Number of Clusters	808
Number of Observations	1500
Number of Observations Used	811
Number of Obs with Nonpositive Weights	689

Data Summary	
Sum of Weights	792.362

Table of colon_screening by sex								
colon_screening	sex	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
.	.	0	.	.	.	.	.	.
	1	5	5.77000	3.48230	0.7282	0.4382	49.8617	18.5927
	2	7	5.80200	2.50761	0.7322	0.3168	50.1383	18.5927
	Total	12	11.57200	4.28310	1.4604	0.5389	100.0000	
1	.	1	0.32500	0.32500	0.0410	0.0411	0.1038	0.1040
	1	144	148.46800	14.41131	18.7374	1.7262	47.4324	3.4030
	2	192	164.21700	13.41812	20.7250	1.6651	52.4638	3.4030
	Total	337	313.01000	18.03524	39.5034	2.0764	100.0000	
2	.	0	.	.	.	.	.	.
	1	199	215.36500	16.63196	27.1801	1.9351	46.0398	2.7810
	2	263	252.41500	16.20571	31.8560	1.9583	53.9602	2.7810
	Total	462	467.78000	20.15730	59.0361	2.0951	100.0000	
Total	.	1	0.32500	0.32500	0.0410	0.0411		
	1	348	369.60300	20.28811	46.6457	2.1294		
	2	462	422.43400	18.32364	53.3133	2.1293		
	Total	811	792.36200	19.16626	100.0000			

Distribution of Accessibility Scores (Tiered E2SFCA)

The SURVEYFREQ Procedure

Data Summary	
Number of Strata	18
Number of Clusters	771
Number of Observations	1500
Number of Observations Used	777
Number of Obs with Nonpositive Weights	723
Sum of Weights	793.858

Table of colon_screening by health_coverage								
colon_screening	health_coverage	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
.	.	1	0.60200	0.60200	0.0758	0.0759	9.0854	9.4550
	employer_plan	3	1.95400	1.13689	0.2461	0.1436	29.4899	18.8565
	medicaid_medicare	2	3.60500	3.13363	0.4541	0.3938	54.4069	24.3915
	not_insured	0	.	.	.	.	.	.
	purchased_plan	1	0.46500	0.46500	0.0586	0.0586	7.0178	7.4414
	something_else	0	.	.	.	.	.	.
	Total	7	6.62600	3.40919	0.8347	0.4285	100.0000	
1	.	0	.	.	.	.	.	.
	employer_plan	166	154.00700	13.56750	19.3998	1.6803	51.2707	3.5631
	medicaid_medicare	112	114.75700	13.07684	14.4556	1.5824	38.2039	3.4423
	not_insured	7	6.84500	3.00947	0.8622	0.3773	2.2788	0.9826
	purchased_plan	25	18.22800	4.39271	2.2961	0.5556	6.0683	1.4574

Table of colon_screening by health_coverage								
colon_screening	health_coverage	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
	something_else	6	6.54300	2.92252	0.8242	0.3679	2.1782	0.9646
	Total	316	300.38000	18.09214	37.8380	2.1050	100.0000	
2	.	16	18.54600	5.60566	2.3362	0.7020	3.8094	1.1339
	employer_plan	240	222.79300	14.70313	28.0646	1.8695	45.7620	2.8100
	medicaid_medicare	91	131.74300	16.18671	16.5953	1.8844	27.0602	2.8316
	not_insured	53	63.45200	10.65479	7.9929	1.3099	13.0331	2.0727
	purchased_plan	41	35.25500	6.53205	4.4410	0.8255	7.2414	1.3338
	something_else	13	15.06300	5.16653	1.8974	0.6467	3.0940	1.0467
	Total	454	486.85200	21.77799	61.3273	2.1215	100.0000	
Total	.	17	19.14800	5.63659	2.4120	0.7059		
	employer_plan	409	378.75400	17.66087	47.7105	2.2006		
	medicaid_medicare	205	250.10500	20.16539	31.5050	2.1894		
	not_insured	60	70.29700	11.03653	8.8551	1.3507		
	purchased_plan	67	53.94800	7.79387	6.7957	0.9892		
	something_else	19	21.60600	5.89727	2.7216	0.7373		
	Total	777	793.85800	21.07053	100.0000			

Distribution of Accessibility Scores (Tiered E2SFCA)

The SURVEYFREQ Procedure

Data Summary	
Number of Strata	18
Number of Clusters	808
Number of Observations	1500
Number of Observations Used	811
Number of Obs with Nonpositive Weights	689
Sum of Weights	792.362

Table of colon_screening by age_grp								
colon_screening	age_grp	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
.	.	0	.	.	.	.	.	.
	18-29	1	0.45700	0.45700	0.0577	0.0577	3.9492	4.0621
	30-49	5	6.40500	3.66064	0.8083	0.4606	55.3491	17.7624
	50-64	4	3.50000	2.01377	0.4417	0.2542	30.2454	15.6661
	65+	2	1.21000	0.86608	0.1527	0.1095	10.4563	7.7242
	Total	12	11.57200	4.28310	1.4604	0.5389	100.0000	
1	.	3	1.91600	1.11491	0.2418	0.1410	0.6121	0.3572
	18-29	9	14.39900	5.63368	1.8172	0.7054	4.6002	1.7510
	30-49	49	37.54000	6.32620	4.7377	0.8018	11.9932	1.9731
	50-64	153	134.86100	12.64309	17.0201	1.5620	43.0852	3.3233
	65+	123	124.29400	13.06542	15.6865	1.5867	39.7093	3.3440
	Total	337	313.01000	18.03524	39.5034	2.0764	100.0000	
2	.	0	.	.	.	.	.	.
	18-29	106	110.71200	12.00005	13.9724	1.4755	23.6675	2.3565
	30-49	208	215.60200	15.92140	27.2100	1.8987	46.0905	2.7782

Table of colon_screening by age_grp								
colon_screening	age_grp	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
	50-64	106	93.98100	10.62892	11.8609	1.3228	20.0909	2.1424
	65+	42	47.48500	8.79343	5.9928	1.0911	10.1511	1.7974
	Total	462	467.78000	20.15730	59.0361	2.0951	100.0000	
Total	.	3	1.91600	1.11491	0.2418	0.1410		
	18-29	116	125.56800	13.11493	15.8473	1.5937		
	30-49	262	259.54700	16.76290	32.7561	1.9851		
	50-64	263	232.34200	15.59325	29.3227	1.8889		
	65+	167	172.98900	15.26979	21.8321	1.8135		
	Total	811	792.36200	19.16626	100.0000			

Distribution of Accessibility Scores (Tiered E2SFCA)

The SURVEYFREQ Procedure

Data Summary	
Number of Strata	18
Number of Clusters	771
Number of Observations	1500
Number of Observations Used	777
Number of Obs with Nonpositive Weights	723
Sum of Weights	793.858

Table of colon_screening by race								
colon_screening	race	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
.	.	0	.	.	.	.	.	.
	1	5	3.06100	1.37854	0.3856	0.1745	46.1968	24.6747
	2	1	0.46500	0.46500	0.0586	0.0586	7.0178	7.4414
	3	1	3.10000	3.10000	0.3905	0.3896	46.7854	27.0420
	4	0	.	.	.	.	.	.
	Total	7	6.62600	3.40919	0.8347	0.4285	100.0000	
1	.	6	7.24500	3.89765	0.9126	0.4892	2.4119	1.2809
	1	253	219.92200	15.43379	27.7029	1.8907	73.2146	3.3154
	2	36	35.47600	7.06940	4.4688	0.8854	11.8104	2.2416
	3	8	14.72900	5.75826	1.8554	0.7194	4.9035	1.8599
	4	13	23.00800	6.58095	2.8983	0.8216	7.6596	2.1071
	Total	316	300.38000	18.09214	37.8380	2.1050	100.0000	
2	.	11	12.29000	4.32684	1.5481	0.5434	2.5244	0.8828
	1	289	253.45300	15.53488	31.9267	1.9462	52.0596	2.8287
	2	66	70.46000	10.23903	8.8756	1.2679	14.4726	2.0088
	3	35	54.37900	10.18869	6.8500	1.2519	11.1695	1.9892
	4	53	96.27000	14.28323	12.1269	1.6990	19.7740	2.6153
	Total	454	486.85200	21.77799	61.3273	2.1215	100.0000	
Total	.	17	19.53500	5.81668	2.4608	0.7276		
	1	547	476.43600	18.30141	60.0153	2.1990		
	2	103	106.40100	12.02581	13.4030	1.4779		
	3	44	72.20800	11.96036	9.0958	1.4510		

Table of colon_screening by race								
colon_screening	race	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
	4	66	119.27800	15.56023	15.0251	1.8287		
	Total	777	793.85800	21.07053	100.0000			

Distribution of Accessibility Scores (Tiered E2SFCA)

The SURVEYFREQ Procedure

Data Summary	
Number of Strata	18
Number of Clusters	801
Number of Observations	1500
Number of Observations Used	806
Number of Obs with Nonpositive Weights	694
Sum of Weights	791.289

Table of colon_screening by code_fips								
colon_screening	code_fips	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
.	3400	0	.	.	.	.	.	.
	3401	0	.	.	.	.	.	.
	3402	0	.	.	.	.	.	.
	3403	0	.	.	.	.	.	.
	34001	1	0.50100	0.50100	0.0633	0.0634	5.0221	5.1917
	34003	2	2.42800	1.85480	0.3068	0.2343	24.3384	16.6335
	34005	0	.	.	.	.	.	.
	34007	1	0.57600	0.57600	0.0728	0.0729	5.7739	5.9223
	34011	1	0.45400	0.45400	0.0574	0.0574	4.5509	4.7219
	34013	1	1.15300	1.15300	0.1457	0.1457	11.5577	11.1842
	34015	0	.	.	.	.	.	.
	34017	2	3.71700	3.23482	0.4697	0.4078	37.2594	22.3726
	34021	0	.	.	.	.	.	.
	34023	0	.	.	.	.	.	.
	34025	0	.	.	.	.	.	.
	34027	0	.	.	.	.	.	.
	34031	1	0.61100	0.61100	0.0772	0.0773	6.1247	6.2612
	34033	0	.	.	.	.	.	.
	34035	0	.	.	.	.	.	.
	34037	1	0.53600	0.53600	0.0677	0.0678	5.3729	5.5358
	34041	0	.	.	.	.	.	.
	Total	10	9.97600	4.08355	1.2607	0.5146	100.0000	
1	3400	4	2.90300	1.47839	0.3669	0.1870	0.9811	0.4985
	3401	12	11.88300	3.05953	1.5017	0.3851	4.0158	1.0204
	3402	18	14.70000	3.93133	1.8577	0.4962	4.9678	1.3015
	3403	15	13.08100	3.40817	1.6531	0.4322	4.4207	1.1343
	34001	6	3.79700	1.88873	0.4798	0.2390	1.2832	0.6373
	34003	36	35.66100	6.39232	4.5067	0.8009	12.0515	2.0479
	34005	22	23.42800	5.65721	2.9607	0.7073	7.9174	1.8255

Table of colon_screening by code_fips								
colon_screening	code_fips	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
	34007	20	15.99400	3.77252	2.0213	0.4779	5.4051	1.2499
	34011	9	6.87800	1.95723	0.8692	0.2481	2.3244	0.6617
	34013	33	30.17500	5.84062	3.8134	0.7350	10.1975	1.8914
	34015	14	11.89400	4.00129	1.5031	0.5051	4.0195	1.3306
	34017	14	14.27300	4.47387	1.8038	0.5641	4.8235	1.4775
	34021	21	11.64000	2.24808	1.4710	0.2870	3.9337	0.7701
	34023	22	20.54400	5.74662	2.5963	0.7206	6.9428	1.8537
	34025	19	21.70700	5.84256	2.7432	0.7281	7.3358	1.8772
	34027	18	15.81100	4.14435	1.9981	0.5229	5.3433	1.3776
	34031	15	17.88500	5.29872	2.2602	0.6670	6.0442	1.7283
	34033	2	2.00100	1.27101	0.2529	0.1606	0.6762	0.4293
	34035	6	7.08200	3.41110	0.8950	0.4299	2.3933	1.1381
	34037	9	9.41900	3.33436	1.1903	0.4213	3.1831	1.1153
	34041	7	5.14900	2.02502	0.6507	0.2558	1.7401	0.6811
	Total	322	295.90500	17.74566	37.3953	2.0836	100.0000	
2	3400	8	7.62300	3.46196	0.9634	0.4348	1.5704	0.7054
	3401	7	6.22300	2.14974	0.7864	0.2723	1.2820	0.4413
	3402	29	24.00400	4.21503	3.0335	0.5336	4.9451	0.8653
	3403	33	37.43000	7.18759	4.7303	0.8920	7.7110	1.4305
	34001	11	11.60400	4.43444	1.4665	0.5574	2.3906	0.9037
	34003	35	33.14500	5.23265	4.1887	0.6655	6.8283	1.0582
	34005	42	37.87000	6.43696	4.7859	0.8040	7.8017	1.2843
	34007	34	33.58500	5.70826	4.2443	0.7117	6.9189	1.1440
	34011	7	7.91000	3.60928	0.9996	0.4540	1.6296	0.7359
	34013	45	45.90500	7.25924	5.8013	0.9005	9.4570	1.4236
	34015	14	15.47700	4.53753	1.9559	0.5723	3.1885	0.9280
	34017	35	41.52000	7.28158	5.2471	0.9088	8.5536	1.4523
	34021	17	13.37100	3.01005	1.6898	0.3806	2.7546	0.6178
	34023	48	53.63800	7.93094	6.7786	0.9766	11.0501	1.5540
	34025	26	21.16200	4.03187	2.6744	0.5122	4.3596	0.8225
	34027	17	12.80500	2.79208	1.6182	0.3554	2.6380	0.5758
	34031	33	49.23300	8.19814	6.2219	1.0037	10.1426	1.5800
	34033	1	0.51300	0.51300	0.0648	0.0649	0.1057	0.1057
	34035	14	15.67500	5.35476	1.9809	0.6720	3.2292	1.0881
	34037	11	12.21100	4.55654	1.5432	0.5726	2.5156	0.9278
	34041	7	4.50400	1.59347	0.5692	0.2018	0.9279	0.3288
	Total	474	485.40800	21.49734	61.3440	2.1006	100.0000	
Total	3400	12	10.52600	3.23767	1.3302	0.4054		
	3401	19	18.10600	2.60350	2.2882	0.3277		
	3402	47	38.70400	4.71297	4.8913	0.5954		
	3403	48	50.51100	7.09686	6.3834	0.8774		
	34001	18	15.90200	4.68612	2.0096	0.5891		
	34003	73	71.23400	6.49361	9.0023	0.8138		
	34005	64	61.29800	7.27583	7.7466	0.8901		
	34007	55	50.15500	5.44557	6.3384	0.6759		
	34011	17	15.24200	3.26610	1.9262	0.4095		
	34013	79	77.23300	7.66529	9.7604	0.9356		
	34015	28	27.37100	5.79724	3.4590	0.7301		

Table of colon_screening by code_fips								
colon_screening	code_fips	Frequency	Weighted Frequency	Std Err of Wgt Freq	Percent	Std Err of Percent	Row Percent	Std Err of Row Percent
	34017	51	59.51000	8.19171	7.5206	1.0110		
	34021	38	25.01100	2.64647	3.1608	0.3401		
	34023	70	74.18200	8.38766	9.3748	1.0117		
	34025	45	42.86900	5.47889	5.4176	0.6747		
	34027	35	28.61600	4.26763	3.6164	0.5415		
	34031	49	67.72900	8.24831	8.5593	0.9919		
	34033	3	2.51400	1.09284	0.3177	0.1381		
	34035	20	22.75700	6.18889	2.8759	0.7741		
	34037	21	22.16600	5.40042	2.8013	0.6776		
	34041	14	9.65300	2.03213	1.2199	0.2575		
	Total	806	791.28900	20.62567	100.0000			

3-Way Interaction: Age Group x Screening Behavior x Insurance Type

The LOGISTIC Procedure

Model Information		
Data Set	WORK.SURVEY_DATA	
Response Variable	colon_screening	colon_screening
Number of Response Levels	3	
Model	cumulative logit	
Optimization Technique	Fisher's scoring	

Number of Observations Read	1500
Number of Observations Used	1500

Response Profile		
Ordered Value	colon_screening	Total Frequency
1	2	893
2	1	593
3	.	14

Probabilities modeled are cumulated over the lower Ordered Values.

Backward Elimination Procedure

Class Level Information						
Class	Value	Design Variables				
age_grp	.	1	0	0	0	
	18-29	-1	-1	-1	-1	
	30-49	0	1	0	0	
	50-64	0	0	1	0	
	65+	0	0	0	1	
insurancetype	.	1	0	0	0	0
	1	-1	-1	-1	-1	-1
	2	0	1	0	0	0
	3	0	0	1	0	0
	4	0	0	0	1	0

Class Level Information						
Class	Value	Design Variables				
	5	0	0	0	0	1
screening_behavior	.	1	0	0	0	
	1	-1	-1	-1	-1	
	2	0	1	0	0	
	3	0	0	1	0	
	4	0	0	0	1	

Step 0. The following effects were entered:

Intercept_2 Intercept_1 age_gr*insura*screen

Model Convergence Status
Quasi-complete separation of data points detected.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
165.1222	78	<.0001

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2161.795	1944.470
SC	2172.422	2369.528
-2 Log L	2157.795	1784.470

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	373.3251	78	<.0001
Score	312.2714	78	<.0001
Wald	271.9988	78	<.0001

Note: No (additional) effects met the 0.05 significance level for removal from the model.

Joint Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_gr*insura*screen	78	271.9988	<.0001

Note: Under full-rank parameterizations, Type 3 effect tests are replaced by joint tests. The joint test for an effect is a test that all the parameters associated with that effect are zero. Such joint tests might not be equivalent to Type 3 effect tests under GLM parameterization.

Note: The following parameters have been set to 0, since the variables are a linear combination of other variables as shown.

age_grp30-49insurancetype4scree4 =	age_grp.insurancetype2screening_ - age_grp.insurancetype2screening4 - age_grp.insurancetype4screening_ + age_grp.insurancetype4screening4 - age_grp30-49insurancetype2screen + age_grp30-49insurancetype2scree4 + age_grp30-49insurancetype4screen
age_grp65+insurancetype5screeni3 =	age_grp.insurancetype.screening_ - age_grp.insurancetype.screening3 - age_grp.insurancetype5screening_ + age_grp.insurancetype5screening3 - age_grp65+insurancetype.screenin + age_grp65+insurancetype.screeni3 + age_grp65+insurancetype5screenin

Analysis of Maximum Likelihood Estimates								
Parameter				DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq

Analysis of Maximum Likelihood Estimates								
Parameter				DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	2			1	0.8432	0.1004	70.5938	<.0001
Intercept	1			1	5.6528	0.3010	352.6372	<.0001
age_gr*insura*screen	.	.	.	1	1.4485	360.8	0.0000	0.9968
age_gr*insura*screen	.	.	2	1	-9.5331	3.6111	6.9693	0.0083
age_gr*insura*screen	.	.	3	1	-2.8220	4.1722	0.4575	0.4988
age_gr*insura*screen	.	.	4	1	8.7175	303.0	0.0008	0.9770
age_gr*insura*screen	.	2	.	1	-11.7988	285.7	0.0017	0.9671
age_gr*insura*screen	.	2	2	1	3.9028	2.1783	3.2103	0.0732
age_gr*insura*screen	.	2	3	1	2.5465	3.2479	0.6147	0.4330
age_gr*insura*screen	.	2	4	1	4.2934	285.6	0.0002	0.9880
age_gr*insura*screen	.	3	.	1	-8.4304	2.3024	13.4068	0.0003
age_gr*insura*screen	.	3	2	1	2.5615	1.0892	5.5303	0.0187
age_gr*insura*screen	.	3	3	1	1.1466	1.3223	0.7519	0.3859
age_gr*insura*screen	.	3	4	1	2.0001	1.9744	1.0262	0.3111
age_gr*insura*screen	.	4	.	1	12.9317	208.4	0.0038	0.9505
age_gr*insura*screen	.	4	2	1	2.8953	2.7425	1.1145	0.2911
age_gr*insura*screen	.	4	3	1	1.7560	2.2604	0.6035	0.4373
age_gr*insura*screen	.	4	4	1	-9.5146	71.5090	0.0177	0.8941
age_gr*insura*screen	.	5	.	1	-3.4671	4.7693	0.5285	0.4673
age_gr*insura*screen	.	5	2	1	-0.7641	1.5687	0.2372	0.6262
age_gr*insura*screen	.	5	3	1	-1.4979	1.8032	0.6900	0.4062
age_gr*insura*screen	.	5	4	1	4.8870	3.3456	2.1337	0.1441
age_gr*insura*screen	30-49	.	.	1	1.8739	339.1	0.0000	0.9956
age_gr*insura*screen	30-49	.	2	1	-1.2605	0.9755	1.6699	0.1963
age_gr*insura*screen	30-49	.	3	1	0.2174	1.6978	0.0164	0.8981
age_gr*insura*screen	30-49	.	4	1	10.4037	276.8	0.0014	0.9700
age_gr*insura*screen	30-49	2	.	1	10.3584	276.9	0.0014	0.9702
age_gr*insura*screen	30-49	2	2	1	0.2241	0.6545	0.1172	0.7321
age_gr*insura*screen	30-49	2	3	1	-0.5086	1.1007	0.2135	0.6440
age_gr*insura*screen	30-49	2	4	1	-10.4378	276.9	0.0014	0.9699
age_gr*insura*screen	30-49	3	.	1	-0.2133	0.7215	0.0874	0.7675
age_gr*insura*screen	30-49	3	2	1	0.3985	0.4620	0.7437	0.3885
age_gr*insura*screen	30-49	3	3	1	0.1820	0.7194	0.0640	0.8002
age_gr*insura*screen	30-49	3	4	1	-0.2133	0.7215	0.0874	0.7675
age_gr*insura*screen	30-49	4	.	1	-9.8043	195.8	0.0025	0.9601
age_gr*insura*screen	30-49	4	2	1	-0.4808	1.0611	0.2053	0.6505
age_gr*insura*screen	30-49	4	3	1	-0.1186	1.5914	0.0056	0.9406
age_gr*insura*screen	30-49	4	4	0	0	.	.	.
age_gr*insura*screen	30-49	5	.	1	-0.7549	0.9296	0.6596	0.4167
age_gr*insura*screen	30-49	5	2	1	0.2557	0.5197	0.2421	0.6227
age_gr*insura*screen	30-49	5	3	1	-0.1719	0.8033	0.0458	0.8305
age_gr*insura*screen	30-49	5	4	1	0.0679	0.8511	0.0064	0.9364
age_gr*insura*screen	50-64	.	.	1	-0.6430	1.8143	0.1256	0.7230
age_gr*insura*screen	50-64	.	2	1	2.5955	1.6626	2.4369	0.1185
age_gr*insura*screen	50-64	.	3	1	-2.0671	2.0466	1.0201	0.3125
age_gr*insura*screen	50-64	.	4	1	-1.4876	1.6250	0.8380	0.3600
age_gr*insura*screen	50-64	2	.	1	0.6024	1.5128	0.1586	0.6905
age_gr*insura*screen	50-64	2	2	1	-1.6526	0.5997	7.5946	0.0059

Analysis of Maximum Likelihood Estimates								
Parameter				DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
age_gr*insura*screen	50-64	2	3	1	0.2673	0.8313	0.1034	0.7478
age_gr*insura*screen	50-64	2	4	1	1.0373	1.3505	0.5900	0.4424
age_gr*insura*screen	50-64	3	.	1	1.6677	1.2723	1.7181	0.1899
age_gr*insura*screen	50-64	3	2	1	-1.3685	0.6543	4.3751	0.0365
age_gr*insura*screen	50-64	3	3	1	1.0221	0.9411	1.1794	0.2775
age_gr*insura*screen	50-64	3	4	1	-0.1570	0.7581	0.0429	0.8359
age_gr*insura*screen	50-64	4	.	1	-0.9802	2.4970	0.1541	0.6947
age_gr*insura*screen	50-64	4	2	1	1.3914	1.2863	1.1702	0.2794
age_gr*insura*screen	50-64	4	3	1	-0.0324	1.3721	0.0006	0.9812
age_gr*insura*screen	50-64	4	4	1	0.5083	1.8485	0.0756	0.7833
age_gr*insura*screen	50-64	5	.	1	-3.0945	2.1569	2.0584	0.1514
age_gr*insura*screen	50-64	5	2	1	0.3926	0.8795	0.1992	0.6553
age_gr*insura*screen	50-64	5	3	1	1.2499	1.3526	0.8539	0.3554
age_gr*insura*screen	50-64	5	4	1	-0.4380	1.1161	0.1540	0.6947
age_gr*insura*screen	65+	.	.	1	-27.4611	6.3557	18.6683	<.0001
age_gr*insura*screen	65+	.	2	1	7.9286	2.9247	7.3491	0.0067
age_gr*insura*screen	65+	.	3	1	6.4487	2.9733	4.7040	0.0301
age_gr*insura*screen	65+	.	4	1	4.7821	4.1638	1.3191	0.2508
age_gr*insura*screen	65+	2	.	1	11.5999	3.9890	8.4562	0.0036
age_gr*insura*screen	65+	2	2	1	-3.2480	1.9022	2.9156	0.0877
age_gr*insura*screen	65+	2	3	1	-3.2480	2.6846	1.4638	0.2263
age_gr*insura*screen	65+	2	4	1	-3.2480	1.9022	2.9156	0.0877
age_gr*insura*screen	65+	3	.	1	7.0366	0.7450	89.2016	<.0001
age_gr*insura*screen	65+	3	2	1	-1.6426	0.2496	43.3251	<.0001
age_gr*insura*screen	65+	3	3	1	-1.8203	0.3650	24.8768	<.0001
age_gr*insura*screen	65+	3	4	1	-1.5693	0.3686	18.1249	<.0001
age_gr*insura*screen	65+	4	.	1	-0.0388	1.8705	0.0004	0.9835
age_gr*insura*screen	65+	4	2	1	-0.0388	1.8705	0.0004	0.9835
age_gr*insura*screen	65+	4	3	1	-1.1943	0.9613	1.5436	0.2141
age_gr*insura*screen	65+	4	4	1	1.3112	1.6998	0.5950	0.4405
age_gr*insura*screen	65+	5	.	1	7.3239	3.9382	3.4586	0.0629
age_gr*insura*screen	65+	5	2	1	-0.8273	1.0012	0.6827	0.4087
age_gr*insura*screen	65+	5	3	0	0	.	.	.
age_gr*insura*screen	65+	5	4	1	-3.2480	2.6846	1.4638	0.2263

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	76.5	Somers' D	0.564
Percent Discordant	20.1	Gamma	0.583
Percent Tied	3.4	Tau-a	0.276
Pairs	550353	c	0.782

Binary Logistic Model with Interaction

The LOGISTIC Procedure

Model Information	
Data Set	WORK.SURVEY_BINARY2
Response Variable	full_adhere

Model Information	
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	1500
Number of Observations Used	1500

Response Profile		
Ordered Value	full_adhere	Total Frequency
1	0	907
2	1	593

Probability modeled is full_adhere=1.

Class Level Information						
Class	Value	Design Variables				
insurancetype	.	1	0	0	0	0
	1	-1	-1	-1	-1	-1
	2	0	1	0	0	0
	3	0	0	1	0	0
	4	0	0	0	1	0
	5	0	0	0	0	1
screening_behavior	.	1	0	0	0	
	1	-1	-1	-1	-1	
	2	0	1	0	0	
	3	0	0	1	0	
	4	0	0	0	1	

Model Convergence Status
Quasi-complete separation of data points detected.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2015.222	1940.031
SC	2020.535	2099.428
-2 Log L	2013.222	1880.031

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	133.1912	29	<.0001
Score	118.3842	29	<.0001
Wald	91.4088	29	<.0001

Joint Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
insurancetype	5	3.4754	0.6271
screening_behavior	4	0.0148	1.0000
insurance*screening_	20	11.5011	0.9322

Note: Under full-rank parameterizations, Type 3 effect tests are replaced by joint tests. The joint test for an effect is a test that all the parameters associated with that effect are zero. Such joint tests might not be equivalent to Type 3 effect tests under GLM parameterization.

Analysis of Maximum Likelihood Estimates							
Parameter			DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept			1	-1.5116	16.8723	0.0080	0.9286
insurancetype	.		1	-6.1887	49.3916	0.0157	0.9003
insurancetype	2		1	3.1658	49.1400	0.0042	0.9486
insurancetype	3		1	1.3685	16.8729	0.0066	0.9354
insurancetype	4		1	3.0217	49.1405	0.0038	0.9510
insurancetype	5		1	-2.3350	26.4913	0.0078	0.9298
screening_behavior	.		1	1.1019	58.8945	0.0004	0.9851
screening_behavior	2		1	0.4257	16.8732	0.0006	0.9799
screening_behavior	3		1	-1.0128	30.6000	0.0011	0.9736
screening_behavior	4		1	-0.9732	26.8170	0.0013	0.9711
insurance*screening_	.	.	1	-4.6686	109.0	0.0018	0.9658
insurance*screening_	.	2	1	4.8767	49.3963	0.0097	0.9214
insurance*screening_	.	3	1	-2.5539	136.9	0.0003	0.9851
insurance*screening_	.	4	1	-2.5935	115.3	0.0005	0.9821
insurance*screening_	2	.	1	8.5321	193.8	0.0019	0.9649
insurance*screening_	2	2	1	-2.9963	49.1408	0.0037	0.9514
insurance*screening_	2	3	1	-1.4300	55.3764	0.0007	0.9794
insurance*screening_	2	4	1	-0.9687	53.3803	0.0003	0.9855
insurance*screening_	3	.	1	-2.8307	58.8966	0.0023	0.9617
insurance*screening_	3	2	1	-0.0454	16.8742	0.0000	0.9979
insurance*screening_	3	3	1	1.6666	30.6008	0.0030	0.9566
insurance*screening_	3	4	1	1.4039	26.8181	0.0027	0.9583
insurance*screening_	4	.	1	8.6762	193.8	0.0020	0.9643
insurance*screening_	4	2	1	-3.0344	49.1426	0.0038	0.9508
insurance*screening_	4	3	1	-0.9028	55.3772	0.0003	0.9870
insurance*screening_	4	4	1	-1.9232	53.3832	0.0013	0.9713
insurance*screening_	5	.	1	-8.5224	100.7	0.0072	0.9326
insurance*screening_	5	2	1	1.5237	26.4933	0.0033	0.9541
insurance*screening_	5	3	1	2.1512	36.7954	0.0034	0.9534
insurance*screening_	5	4	1	3.4334	33.7123	0.0104	0.9189

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	59.5	Somers' D	0.308
Percent Discordant	28.7	Gamma	0.349
Percent Tied	11.7	Tau-a	0.147
Pairs	537851	c	0.654

Model Fit Comparison with AIC and -2LL

The LOGISTIC Procedure

Model Information	
Data Set	WORK.SURVEY_BINARY2
Response Variable	full_adhere
Number of Response Levels	2

Model Information	
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	1500
Number of Observations Used	1500

Response Profile		
Ordered Value	full_adhere	Total Frequency
1	0	907
2	1	593

Probability modeled is full_adhere=1.

Stepwise Selection Procedure

Class Level Information						
Class	Value	Design Variables				
insurancetype	.	1	0	0	0	0
	1	0	1	0	0	0
	2	0	0	1	0	0
	3	0	0	0	1	0
	4	0	0	0	0	1
	5	-1	-1	-1	-1	-1
screening_behavior	.	1	0	0	0	
	1	0	1	0	0	
	2	0	0	1	0	
	3	0	0	0	1	
	4	-1	-1	-1	-1	

Step 0. Intercept entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

-2 Log L	=	2013.222

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
118.3842	29	<.0001

Step 1. Effect insurancetype entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2015.222	1918.918
SC	2020.535	1950.797
-2 Log L	2013.222	1906.918

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	106.3042	5	<.0001
Score	95.8083	5	<.0001
Wald	80.9484	5	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
24.5931	24	0.4281

Note: No effects for the model in Step 1 are removed.

Note: No (additional) effects met the 0.05 significance level for entry into the model.

Summary of Stepwise Selection								
	Effect							
Step	Entered	Removed	DF	Number In	Score Chi-Square	Wald Chi-Square	Pr > ChiSq	Variable Label
1	insurancetype		5	1	95.8083		<.0001	insurancetype

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
insurancetype	5	80.9484	<.0001

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-1.1073	0.1480	55.9675	<.0001
insurancetype	.	1	-1.6332	0.6137	7.0819	0.0078
insurancetype	1	1	0.6881	0.1595	18.6125	<.0001
insurancetype	2	1	0.2296	0.2178	1.1116	0.2917
insurancetype	3	1	1.2794	0.1694	57.0540	<.0001
insurancetype	4	1	0.3341	0.3211	1.0827	0.2981

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
insurancetype . vs 5	0.479	0.103	2.225
insurancetype 1 vs 5	4.885	2.746	8.689
insurancetype 2 vs 5	3.088	1.569	6.078
insurancetype 3 vs 5	8.824	4.882	15.950
insurancetype 4 vs 5	3.429	1.418	8.288

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	45.1	Somers' D	0.260
Percent Discordant	19.0	Gamma	0.406
Percent Tied	35.9	Tau-a	0.125
Pairs	537851	c	0.630